

# Unpack a Suitcase SOP

One episode empties one suitcase. The zip is drawn open, the lid falls back onto its rest, three folded garments go onto three spots on the tabletop sorted by type, three toiletry bottles go into the toiletry tray, one pair of shoes sits on the mat, and the pouch is nested inside the tote. The case is empty and the lid is still open when the episode ends.

The order never changes: zip, lid, garments, toiletries, shoes, bags. A zone is finished before the next one starts, and nothing already placed is touched again.

The right gripper draws the zip, opens the lid, moves every garment, and moves both bags. The left gripper anchors the case for the zip and the lid, moves the toiletries and the shoes to their zones on the left, and gapes the tote for the nest. No garment ever touches the left gripper and the right gripper never reaches the toiletry tray or the mat.

The case never leaves its cradle, nothing is turned in the air, no item is pulled out from under another, and no gripper opens anywhere but over the place the item is going.

## Setup

Complete both checklists before starting an episode.

### Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the case in its cradle at the back, the three garment spots down the right, the toiletry tray and the shoe mat down the left, and the bag spot at the front center.
3. The whole zip track is in frame end to end, and the case interior is visible from above so every lift can be seen leaving it.
4. Both arms are at home with grippers open.
5. The tabletop is clear of anything but the zones listed below.
6. Nothing obstructs the paths between the case and the four destination zones, and the left arm rests on the case's left side panel while the right arm reaches both ends of the zip track.

### Materials checklist

1. One suitcase lying flat in the cradle, base on the cradle floor, lid closed and facing up.
2. The zip is a single straight track running left to right across the front edge of the case, with one slider parked at the **left end stop**, carrying a fabric loop tab big enough for a gripper to pinch.
3. The lid is hinged along the back edge and carries a **lift loop** stitched at the middle of its front edge, standing clear of the shell so a gripper can pinch it from above.

4. Everything inside the case lies in one layer, nothing resting on top of anything else, with a clear gap between items so a gripper can close on one without touching its neighbors.
5. The right half of the case holds three folded garments lying face up, one of each type: one shirt, one pair of pants, and one sweater, each folded into a flat rectangle with a clean folded edge a gripper can close across.
6. The left half of the case holds three toiletry bottles with their caps closed, in a line running near to far, and one pair of shoes sitting sole down, toes toward the front edge, side by side and not touching.
7. The front center of the case holds one tote that keeps its shape when set down, empty, with an open mouth, no fastening, and a single carry handle across the mouth; and one small pouch lying flat and empty beside it.
8. The three garment spots, the toiletry tray, the shoe mat, and the bag spot are all empty.

## Workspace layout

- **Case cradle:** at the back of the tabletop, centered between the two arms, with low walls that hold the case square. The case stays in it for the whole episode and no gripper ever has to hold it still.
- **Lid rest:** a low block behind the cradle. The open lid falls back onto it and stays clear of the case mouth for the rest of the episode.
- **Garment spots:** three clear areas of bare tabletop in a row down the right side, within reach of the right arm. They are not marked. The **shirt spot** is nearest the case, the **pants spot** sits beyond it, and the **sweater spot** is nearest the front edge, far enough apart that no garment touches another.
- **Toiletry tray:** on the left, within reach of the left arm. One open tray with a flat floor and low walls, no slots and no dividers, big enough to hold all three bottles at once.
- **Shoe mat:** at the front left, a flat mat marked into a left half and a right half.
- **Bag spot:** a marked square at the front center, reachable by both arms.

The contents are packed on the side they leave toward: garments in the right half of the case for the right arm, toiletries and shoes in the left half for the left arm. No carried item crosses the midline and no carried item passes over another item.

## Vocabulary

- **Arm assignment:** the right arm works the case's right half, the zip, the lid, and the garment spots. The left arm anchors, works the case's left half, the toiletry tray, and the mat, and gapes the tote. Neither arm takes the other's zones.
- **Cradle:** the walled tray holding the case. It does the holding, so no gripper ever grips, steadies, or lifts the case.

- **Track:** the line of teeth the slider runs along. It has a **left end stop**, where the slider parks while the case is closed, and a **right end stop**, where it parks once the case is open.
- **Loop tab:** the fabric loop on the slider. It is the only part of a zip a gripper ever touches.
- **Lift loop:** the loop on the front edge of the lid. It is the only part of the lid a gripper ever touches.
- **Anchor:** the closed left gripper pressed onto the case's **left side panel**, holding the case square in the cradle while the right gripper draws or lifts. An anchor never travels and never sits on the front face where the slider runs.
- **Draw:** the right gripper pinching the loop tab and pulling the slider along the track, left end stop to right end stop, in one continuous motion along the line of the track.
- **Lift:** taking an item straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Load:** carrying an item level to its place, lowering it straight down until it is resting, and releasing.
- **Gape:** the left gripper pinching the left end of the tote's mouth rim and pulling it out and down so the mouth stands open. A gape is held for the whole step, not pulled and released.
- **Nest:** the pouch resting on the floor of the tote with no part of it standing at the mouth.
- **Garment types:** the garments are named **shirt**, **pants**, and **sweater**, never by color or pattern. There is one garment of each type, and each type has one spot and only that spot.
- **Settled:** the item stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or falling over.
- **Release point:** over the garment spot, the tray, the mat half, or the bag the item is going into, once the item is already resting. A gripper opens nowhere else.

## Steps

Steps 1 to 6 are one suitcase. Every item is lifted straight up out of the case and carried in the orientation it came out in.

### Step 1: Draw the zip open

**Goal:** the zip open end stop to end stop, with the case still square in its cradle.

- The **left gripper**, closed, anchors the case on its **left side panel**, level with the middle of the case, and holds it there for the whole step.
- The **right gripper** comes down onto the **loop tab** from above the track, where the slider is parked at the left end stop, and pinches it.
- Draw the slider straight along the track, left to right, in one continuous motion, until it reaches the right end stop and will not travel further. Release the tab and lift clear.
- The **left gripper** lifts clear only once the right gripper is off the track.

The draw pulls the case rightward, and the anchor is the only thing keeping it square. The anchor stays on the side panel so the two grippers never meet on the front of the case.

**Check:** the slider sits at the right end stop, the track is open along its whole length, and the case has not shifted or lifted in the cradle. If the slider stopped short, take the loop tab again and draw it the rest of the way.

## Step 2: Open the lid

**Goal:** the lid lying back on the lid rest, with the whole case interior open to view.

- The **left gripper** anchors the case on its **left side panel** again and holds it there for the whole step.
- The **right gripper** comes down onto the **lift loop** at the middle of the lid's front edge and pinches it.
- Raise the loop up and back over the hinge in one continuous motion until the lid passes upright and falls back onto the lid rest. Release the loop and lift clear.
- The **left gripper** lifts clear once the lid is resting.

Follow the lid with the loop rather than pulling against the hinge. The loop is flexible, so the gripper never has to turn as the lid swings.

**Check:** the lid rests on the lid rest, the case mouth is clear from front to back, and the case has not lifted out of the cradle. If the lid has fallen back toward the case, take the lift loop again and carry it the rest of the way onto the rest.

## Step 3: Lay the garments out on the tabletop

**Goal:** the case's right half empty and each garment lying flat on the spot for its type.

Work by type in this order: the shirt, then the pants, then the sweater.

- The **right gripper** closes across the **folded edge** of the garment at its middle, taking the full thickness of the fold, and lifts it straight up out of the case.
- Carry it level to its spot: the shirt to the **shirt spot** nearest the case, the pants to the **pants spot** beyond it, the sweater to the **sweater spot** nearest the front edge.
- Lower it flat onto the bare tabletop, squared up the way it came out, and release once it is resting.
- Move to the next type only once the current garment is down on its spot.

Take each garment straight up. Never slide one out from under or between its neighbors, never shake a garment out to check what it is, and never fold or straighten one once it is down.

**Check:** the case's right half is empty, the three garments lie flat on the tabletop in the order shirt, then pants, then sweater running from the case to the front edge, and no garment touches

another or overlaps one. If a garment lands on the wrong spot or on top of another, lift it clear and lower it into place again.

#### **Step 4: Move the toiletries into the tray**

**Goal:** all three bottles resting in the toiletry tray and the tray holding nothing else.

- The **left gripper** pinches the bottle nearest the front edge around its body, below the cap and above the base, and lifts it straight up out of the case.
- Carry it level to the **toiletry tray**, hold it inside the tray walls, lower it straight down until it rests on the tray floor, and release.
- Take the next bottle in the line the same way, then the last one, each into a clear part of the tray floor beside the ones already there.

Work the bottles near to far, the order they are packed in. Do not sort them by size and do not go back to a bottle once you have moved on.

**Check:** all three bottles rest in the tray, none is on a tray wall, none hangs over the rim, none is stacked on another, all are settled, and the case's left half holds only the shoes. If a bottle sits on the rim or on top of another, lift it clear and lower it into a free part of the floor.

#### **Step 5: Set the shoes on the mat**

**Goal:** both shoes sitting sole down on the mat, toes toward the front edge, side by side and not touching.

- The **left gripper** closes on the **left shoe** across its heel, taking both sides of the heel counter, and lifts it straight up out of the case, level and sole down.
- Carry it to the **left half of the mat**, lower it until the whole sole is resting flat, and release.
- Take the **right shoe** the same way onto the **right half of the mat**.

Both shoes are already packed sole down and facing the front edge, so neither one is turned at any point. Carry each one the way it came out.

**Check:** both shoes sit flat on the mat, soles down, toes toward the front edge, one on each half, side by side and not touching, and the case's left half is empty. If a shoe lands on its side or overhangs the mat, lift it clear and set it down again.

#### **Step 6: Nest the pouch in the tote**

**Goal:** the tote standing on the bag spot with the pouch resting inside it.

- The **right gripper** pinches the **carry handle** at its middle, lifts the tote straight up out of the case, carries it level to the **bag spot**, lowers it until its base rests flat on the mark, and releases.

- The **left gripper** gapes the tote: it pinches the **left end of the mouth rim**, pulls it out and down until the mouth stands open, and holds it there for the rest of the step.
- The **right gripper** pinches the pouch across its middle, lifts it straight up out of the case, carries it level over the open mouth, lowers it straight down until it rests on the tote floor, and releases.
- Both grippers lift clear and the mouth falls shut.

The tote is set down before it is opened, so nothing is held in the air while it is worked on.

**Check:** the tote stands on the bag spot with its base flat, the pouch is fully inside with no part of it at the mouth, and the case is empty. If the pouch stands proud of the mouth, lift it out, gape the tote again, and lower it back in.

### Step 7: End the episode

- Confirm the case is empty with its lid back on the lid rest, the three garments lie flat on their spots, the three bottles rest in the tray, both shoes sit on the mat, and the pouch is inside the tote.
- Return both arms home, then stop recording.

## After the episode: reset the workspace

This reset is not recorded.

1. Lift the pouch out of the tote and lay both bags flat in the front center of the case, tote mouth toward the front edge.
2. Set both shoes back in the case's left half, sole down, toes toward the front edge, side by side and not touching.
3. Take the three bottles out of the tray and set them back in the case's left half in a line running near to far, caps closed. Leave the tray empty.
4. Lay the three garments back in the case's right half in one layer, face up, with a clear gap between them. Refold any garment that has come loose so it is a flat rectangle with a clean folded edge.
5. Vary which garment sits in which position in the case between episodes rather than rebuilding the same layout every time. Nothing rests on top of anything else.
6. Lift the lid off the rest, lay it down onto the case, and draw the slider back to the left end stop so the case is closed.
7. Square the case in the cradle: base flat on the cradle floor, lid facing up.
8. Check the loop tab and the lift loop are attached and the zip runs freely end to end. A zip that is starting to catch is replaced before the next episode.
9. Replace any item that is damaged: a case that will not sit square, a garment that will not hold a fold, a leaking bottle, a shoe that will not sit flat, or a tote that no longer holds its

shape.

10. Wipe the tabletop clear of the three garment spots and confirm the cradle, the lid rest, the tray, the mat, and the bag spot sit firm and have not shifted.
11. Run both Setup checklists again.

## SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

### How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

### Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

## Violations

### Violation: Wrong order

- **Visible cue:** the lid is lifted before the zip reaches the right end stop, an item is taken out before the lid is on its rest, a toiletry or a shoe moves before all three garments are down on their spots, or the bags move before the shoes are on the mat.
- **SOP rule broken:** Steps 1 to 6, the order never changes and a zone is finished before the next one starts.
- **Coaching note:** finish the zone you are in before you start another.

### Violation: Arms swapped roles

- **Visible cue:** the left gripper takes a garment, the zip, the lid, or a bag; or the right gripper reaches the toiletry tray, the mat, or the case's left half.
- **SOP rule broken:** Steps 1 to 6, the right arm works the zip, the lid, the case's right half, and the garment spots, and the left arm works the case's left half, the tray, and the mat.
- **Coaching note:** each arm stays on its own zones. Nothing crosses the middle.

### Violation: Zip drawn without an anchor

- **Visible cue:** the left gripper is off the case while the zip is drawn, or it sits on the front face on or beside the track instead of on the left side panel, or the case twists, shifts, or is pulled out of the cradle as the slider travels.
- **SOP rule broken:** Steps 1 and 2, the case is anchored on its left side panel for the whole draw and the two grippers never share the front face.

- **Coaching note:** anchor first, then pull. The anchor belongs on the side of the case, never on the zip line.

#### **Violation: Zip drawn wrong or left short**

- **Visible cue:** the slider is pulled forward off the case, lifted off the fabric, drawn in an arc rather than along the teeth, or drawn right to left; or the draw finishes short of the right end stop with the case still fastened along part of the track and the episode moves on to the lid.
- **SOP rule broken:** Step 1, one continuous motion along the line of the track, left end stop to right end stop.
- **Coaching note:** keep the tab moving along the teeth, and look at the stop before you move on.

#### **Violation: Lid forced or dropped**

- **Visible cue:** the lid is pushed, shoved, or levered open by the shell instead of raised by the lift loop; the right gripper releases it before it passes upright and it falls back onto the case; or it lands beside the lid rest and is left there.
- **SOP rule broken:** Step 2, the right gripper raises the lift loop up and back in one continuous motion until the lid falls onto the lid rest.
- **Coaching note:** carry the loop the whole way. Let go once the lid is resting on the block.

#### **Violation: Case lifted or moved**

- **Visible cue:** the case leaves the cradle, or either gripper lifts, drags, rotates, tips, or holds it still by closing on it at any point.
- **SOP rule broken:** Steps 1 to 6, the case never leaves its cradle and the cradle does the holding.
- **Coaching note:** the cradle does that job. If the case is moving, stop and reseal it.

#### **Violation: Item dug out from under another**

- **Visible cue:** the right gripper works a garment or a bag out from beneath or between its neighbors, or the left gripper works a bottle or a shoe out the same way, or either gripper moves one item aside to reach another.
- **SOP rule broken:** Step 3, no item is pulled out from under another and each item is taken straight up.
- **Coaching note:** everything is in one layer already. Come down on it and lift straight up.

#### **Violation: Item taken by the wrong part**

- **Visible cue:** the right gripper takes a garment by a loose corner, a sleeve, or a single leaf of the fold rather than across the folded edge, takes the tote by its body rather than its carry handle, or takes a zip or a lid by anything but its loop; or the left gripper takes a bottle by its cap or a shoe by its laces, tongue, or sole.

- **SOP rule broken:** Steps 1 to 6, each item has one grasp.
- **Coaching note:** land on the grasp point or reset and take it again.

#### **Violation: Item dragged instead of lifted**

- **Visible cue:** an item slides along the case floor or the tabletop still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 1 to 6, the gripper lifts straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

#### **Violation: Item turned or regripped in the air**

- **Visible cue:** the right gripper flips, swings, or shakes out a garment during the carry or turns a bag over; the left gripper rolls a shoe off its sole; or either gripper shifts, rolls, or seats its hold again on an item without setting it down first.
- **SOP rule broken:** Steps 1 to 6, nothing is turned in the air, each item is carried in the orientation it came out in, and it is set down before a new hold is taken.
- **Coaching note:** orientation is set at the pick, never in the air. Put it down before you take a new hold.

#### **Violation: Garment on the wrong spot or not flat**

- **Visible cue:** the right gripper lays the shirt, the pants, or the sweater on another type's spot, lays two garments on one spot or touching, works the types in any order other than shirt, then pants, then sweater, or leaves a garment folded over on itself or on top of another.
- **SOP rule broken:** Step 3, each garment type has one spot, the types are worked in a fixed order, and each garment is lowered flat with no garment touching another.
- **Coaching note:** name the type before you lift it, then lay it flat on its own spot.

#### **Violation: Toiletry not in the tray**

- **Visible cue:** the left gripper leaves a bottle on a tray wall, hanging over the rim, on top of another bottle, or on the tabletop beside the tray; or it works the bottles in any order other than near to far.
- **SOP rule broken:** Step 4, all three bottles rest on the tray floor and the bottles are worked near to far.
- **Coaching note:** inside the walls, down on the floor, next to the ones already there.

#### **Violation: Shoes wrong side or not flat**

- **Visible cue:** the left gripper puts the left shoe on the right half of the mat or the reverse, sets a shoe down on its side or on its toe, leaves the toes pointing away from the front edge, leaves the two shoes touching, or leaves a shoe overhanging the mat.

- **SOP rule broken:** Step 5, both shoes sole down, toes toward the front edge, one on each half of the mat, side by side and not touching.
- **Coaching note:** left shoe left half, right shoe right half, both flat.

#### **Violation: Bag loaded without a gape**

- **Visible cue:** the right gripper brings the pouch to a tote whose mouth is shut or drives it against the tote's side; or the left gripper takes the gape only after the pouch is in the air, or comes off the rim before the pouch is resting.
- **SOP rule broken:** Step 6, the left gripper holds the mouth open before the pouch is picked up and keeps holding until it rests on the tote floor.
- **Coaching note:** gape first, let go last.

#### **Violation: Item not lowered onto its place**

- **Visible cue:** a gripper opens above a garment spot, the tray, the mat, or the tote mouth and the item falls the rest of the way; a gripper pushes an item in sideways instead of lowering it; a gripper presses, pats, or squashes a garment, the contents of the tote, or the items still in the case; a gripper nudges or slides an item into position after releasing it; or a gripper opens over the case, over another zone, or over open space.
- **SOP rule broken:** Steps 3 to 6, a gripper opens over the place the item is going into, only once the item is resting, and a bad placement is corrected by lifting the item clear and lowering it again.
- **Coaching note:** take it all the way down before you open the gripper, then leave it how it lands or lift it and lay it again.

#### **Violation: Item dropped or something knocked over**

- **Visible cue:** an item falls onto the tabletop or the floor, either gripper strikes the case or the cradle and it shifts, the tray or the mat is pushed out of its spot, a bottle is knocked over, the lid is knocked off its rest, a garment already down is dragged out of place, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, each item stays on a clear travel path, nothing already placed is touched again, and a gripper releases only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

#### **Violation: Wrong episode ending**

- **Visible cue:** the episode ends with something still in the case, a garment spot empty, a bottle outside the tray, a shoe off the mat, the pouch outside the tote, or either arm away from home.
- **SOP rule broken:** Step 7, confirm the empty case and all four loaded zones, return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

## Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Zip jammed or slider derailed** under a straight draw along the track. Replace the case before the next episode.
- **Loop tab or lift loop missing or torn off** when the step begins.
- **Lid will not stay on the lid rest** and falls back onto the case under its own weight. Reseat or replace the rest before the next episode.
- **Defective item:** a garment that comes apart in the gripper and will not hold a fold, a leaking or capless bottle, a shoe that will not sit flat, or a tote that collapses when set down. Replace it before the next episode.
- **Cradle, lid rest, tray, mat, or bag spot shifts out of its spot** during the episode.
- **The three bottles will not all fit** on the floor of the tray.

## Annotation subtasks (from SOP)

1. Anchor the case for a zip or a lid
2. Draw a zip open
3. Open the case lid onto the lid rest
4. Move one garment from the case onto its spot on the tabletop
5. Move one toiletry bottle into the tray
6. Set one shoe on the mat
7. Move the tote from the case to the bag spot
8. Hold the tote mouth open
9. Nest the pouch inside the tote
10. Return both arms home and end the episode